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Overview

FPGA Accelerated Spatial Tracking in Augmented Reality (FAST-AR)

Objective: Accelerate a gesture recognition model to decrease latency for augmented reality controls.

Architecture: Heterogenous System-on-Chip (**SoC**) co-design using both programmable logic (**FPGA/PL**) and a processing system (SoC/PS) for gesture acceleration, supported by an external SoC for video pipeline.

Hardware Contribution: Hardware frame preprocessor to reduce SoC/PS load by reducing problem size, handling all interfacing and downscaling.

Demo: Controlling a rover via hand gestures to demonstrate the effectiveness of a hardware accelerated gesture recognition model.

Additionally, we demonstrate what the hardware is doing by displaying image captured by the camera and the region of interest (**ROI**) that the downscaler operates on.

Hardware Pipeline (PL)

Video in and Demosaic

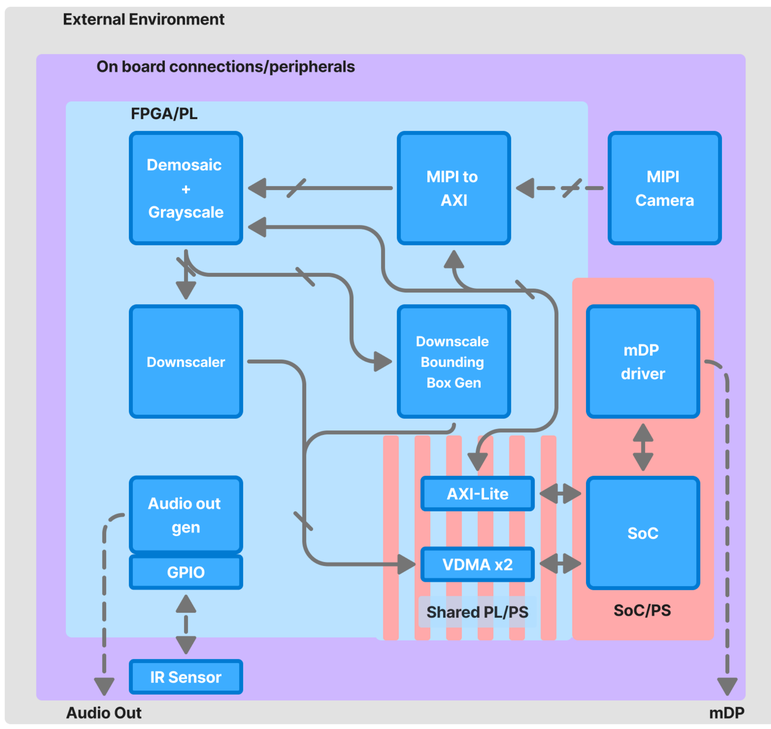
- **Camera data** over 2-lane CSI as RAW10.
- **RAW10** data truncated to **RAW8** and **demosaiced**.
- Data streams out as **RGB888** over **AXI4-Stream.**

Primary Pipeline

- RGB data is **downscaled** from **1080p** to **224x224.**
- Downscaled frame is pushed to **Video Direct Memory Access (VDMA)**.

Secondary/Demo Pipeline

- Split stream is **grayscaled**.
- A **bounding box** is drawn over the downscaling **region**.
- Bound image sent to **VDMA.**



Challenges

PYNQ Ecosystem

- MediaPipe for the model required internet, but the AUP lacks WiFi or ethernet.
- Long flash times and opacity of the FPGA lead to long software deployments.
- PYNQ's Jupyter notebooks often disconnect mid program.

Custom IP

- Properly complying with the AXI4-Stream protocol is difficult!
- Cameras are prone to dropping pixels meaning all pipeline hardware must be robust and able to recover based on AXI4-Stream handshake signals.

IP Configuration and Documentation

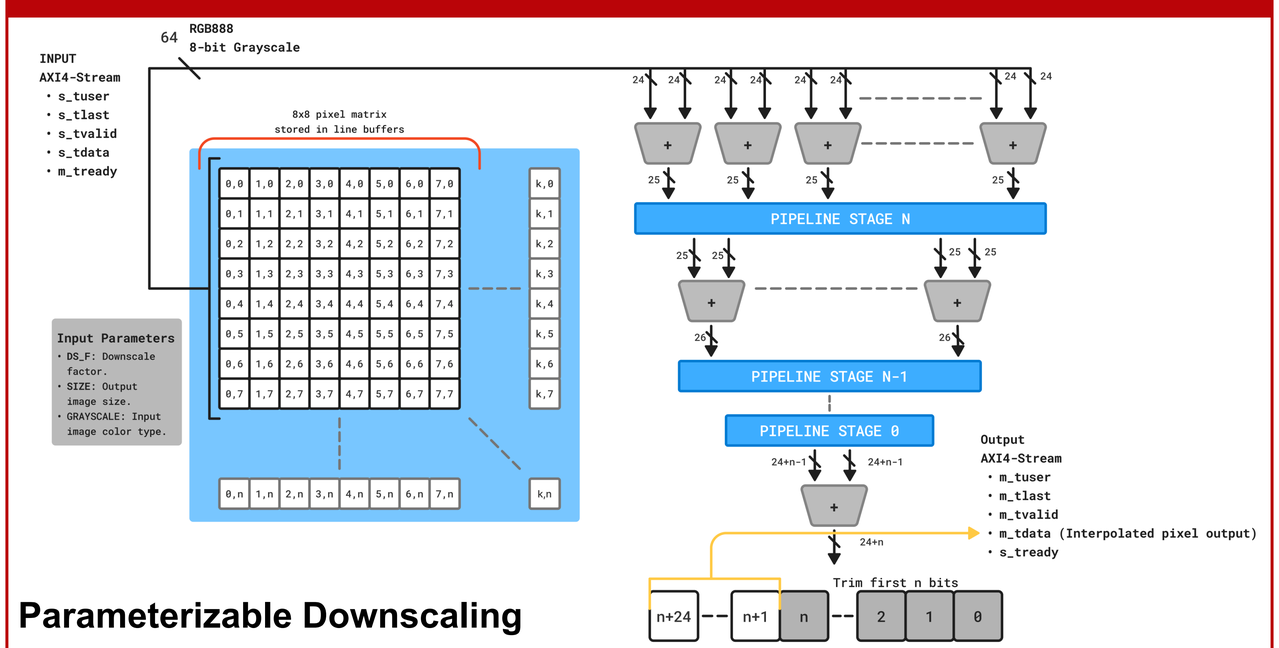
- Vivado/Xilinx IP configuration requires large time sink to both configure properly in hardware and then configure properly over an AXI-lite bus.

Hardware Failures *and the AUP-ZU3…*

- AUP-ZU3 boards have poor ESD protection and often fail short.
- PCAM5 modules are also sensitive and prone to ESD events.

Software Pipeline (PS)

Gesture Acceleration Hardware



Parameterizable Downscaling

- Streams a 1080p frame and does NxN interpolation taking about 76.8 µS* per frame.
- Uses an auto-pipelining** adder tree to increase odds of meeting timing without designer intervention. No DSP slices are used due to tree design and bit manipulation!
- Operates on both RGB888 data or grayscale data depending on model type. Grayscale operation gives a 2/3 reduction in FPGA resources utilized.

Grayscale

- A module that converts a full color RGB image to a single, averaged channel (8 bit).
- Very useful for resource or compute constrained systems due to 2/3 reduction in data.

* *76.8 µS is the time to downscale one 1080p frame to 224x224 before it is sent to VDMA/Mapped Main Memory assuming operation at 200MHz with no stalls and does not account for blanking time of the frame.*
** *Automatic pipelining done at instantiation or synthesis.*

Future Work

Deep Processing Unit (DPU) accelerated Gesture Recognition

- DPU is a proprietary RISC core(s) on FPGA fabric that runs the recognition model.
- Zero software overhead but increases HW complexity and configuration complexity.

High resolution (4K) global shutter camera

- Global shutter removes motion blur which gives higher quality images to the model.
- Higher resolution gives more data, allowing for better downscaled images.

Hardware stereoscopic video pipeline (HW SBS)

- A HW video pipeline reduces software overhead immensely and reduces camera to display latency by an order of magnitude. Below are SWAGs of latency of each.
  - **HW SWAG →** ~3000 cycles @ 200MHz **≈ 15 µS** (as low as **100 nS** for raw feed)
  - **SW SWAG →** 2 DRAM R/W + SW processing **≈ 25 mS** (conservative)

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